DATA SCIENCE 2025

# Evaluating the Role of Financial News and Social Media Sentiment in Enhancing Technical Indicator-based Stock Price Movement Prediction with XGBoost

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# Overview

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# Rationale

"Despite significant advancements in stock price prediction, achieving consistently accurate results remains a major challenge due to the complex and volatile nature of financial markets. Traditional models that rely solely on technical indicators often overlook real-time factors like public sentiment or breaking news. With recent developments in machine learning and NLP, particularly models like FinBERT, there’s a growing opportunity to integrate sentiment from financial news and social media with technical data. This project addresses the lack of unified frameworks that combine both elements by developing a hybrid model using FinBERT for sentiment analysis and XGBoost for classification. The goal is to enhance prediction performance and demonstrate the real-world value of sentiment-aware forecasting."



# Aim

- Develop and evaluate a sentiment-enhanced stock price movement prediction model by integrating social media and financial news sentiment into an XGBoost classifier
- Assess whether sentiment-based features (extracted using FinBERT through some transformation methods) can improve model's prediction performance

# Objectives

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- Data Collection & Preparation: Gather and process stock, news, and Twitter data.
- Feature Engineering: Generate Technical Indicators and use FinBERT to extract sentiment from text.
- Model Development & Evaluation: Build XGBoost models with and without sentiment features and compare their performance.
- Performance Optimization: Apply transformations (e.g., standardization, dimensionality reduction) to improve results.
- Interpretability & Deployment: Use SHAP for feature importance and consider real-world deployment challenges.

# System Overview

DATA SOURCES → PREPROCESSING → FEATURE ENGINEERING

↓

EXPLAINABILITY ← EVALUATION ← MODELLING

# Data Sources

HISTORICAL QUANTITATIVE

SOCIAL MEDIA


FINANCIAL NEWS

# Preprocessing

- Remove irrelevant columns
- Expand date columns
- Drop original date column

- Aggregate by date
- Reduce duplicated rows


- Use raw contents (original data format already sufficient)

Merge all components into 1 based on date

# Feature Engineering

- Expand technical indicators from historical price & volume data

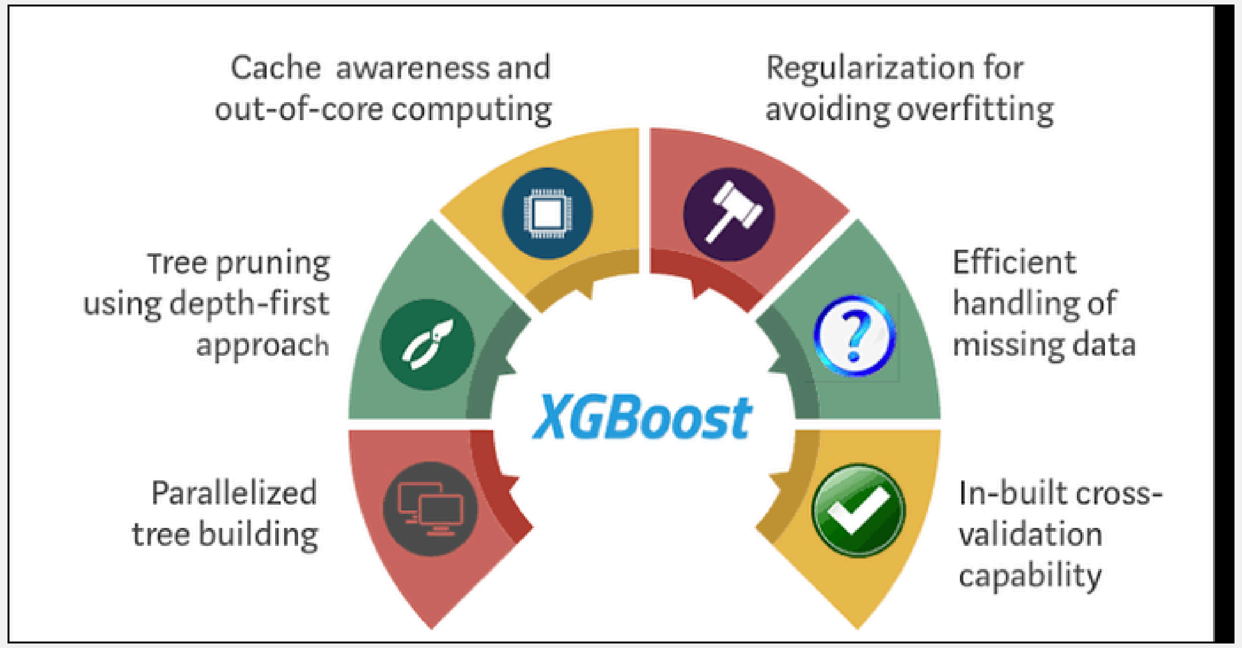
- Clean
- Vectorise (FinBERT)
- Sentiment classification (FinBERT)


- Clean
- Vectorise (FinBERT)
- Sentiment classification (FinBERT)
- Standardise + UMAP (later on to maximise value)

# Modeling

$$Y_{t+1} = \begin{cases} 0, & O_{t+1} \leq C_t \\ 1, & O_{t+1} > C_t \end{cases}$$

BINARY CLASSIFICATION SETUP




XGBOOST ALGORITHM

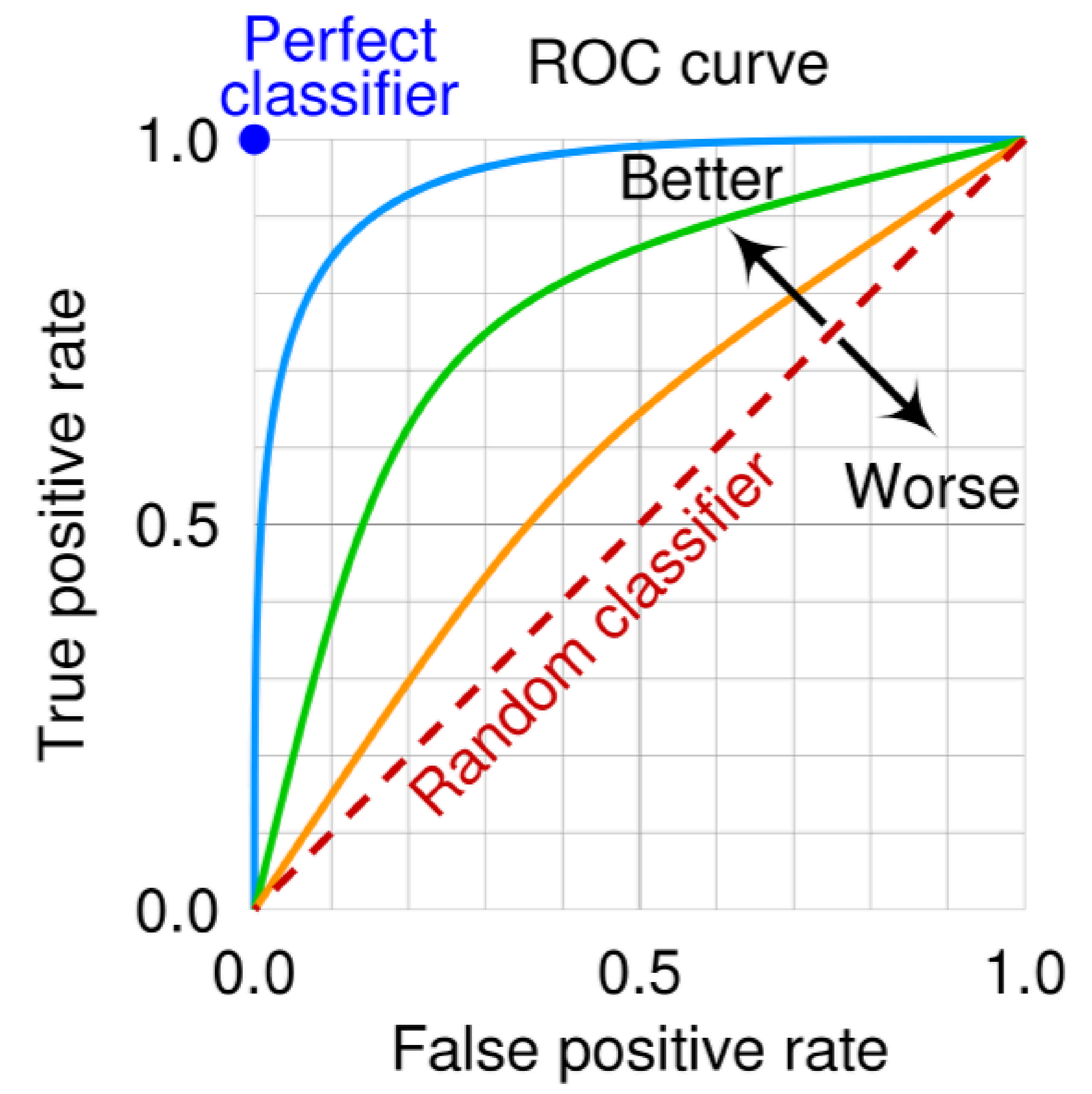
# Evaluation

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

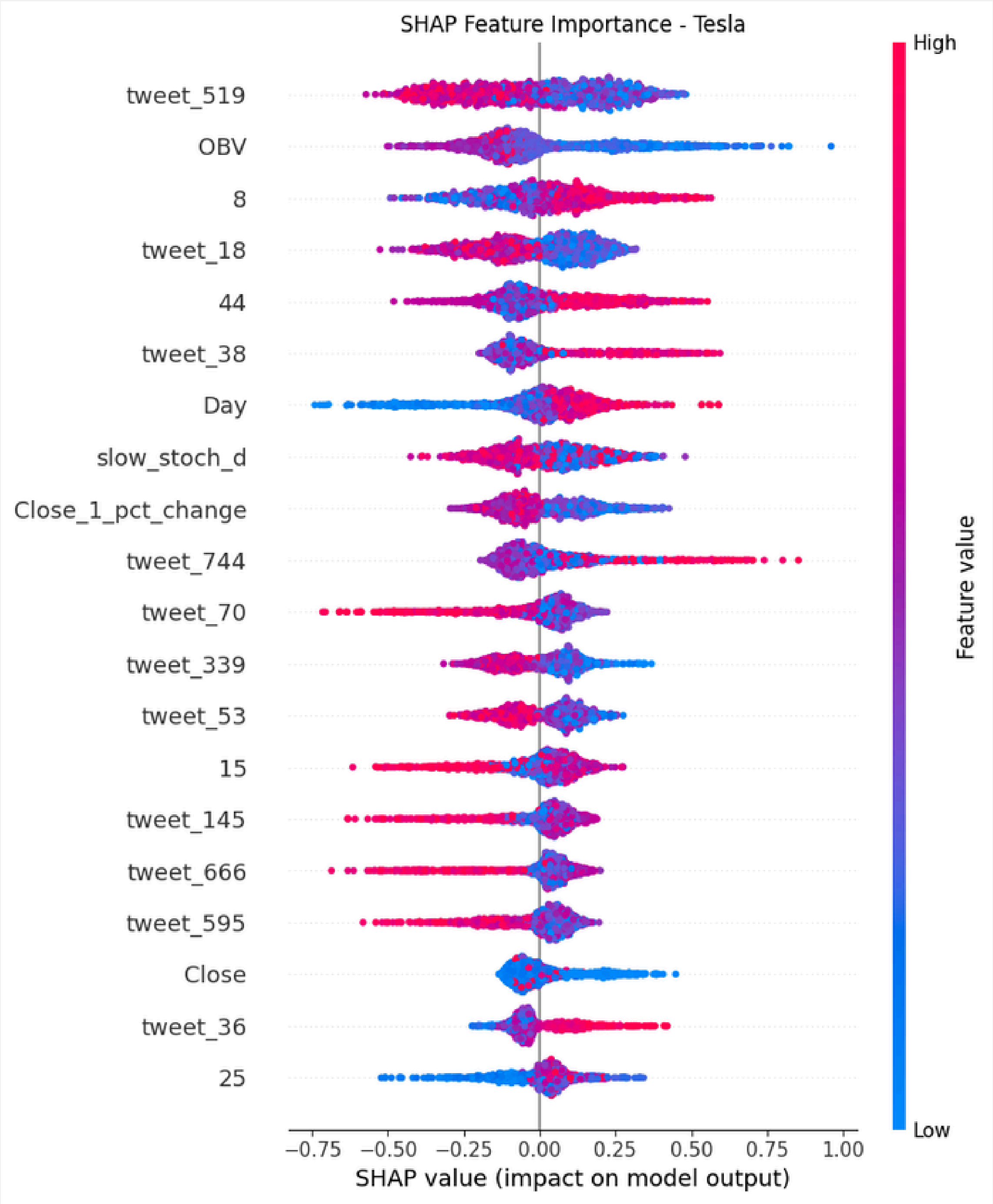
$$Precision = \frac{TP}{TP + FP}$$

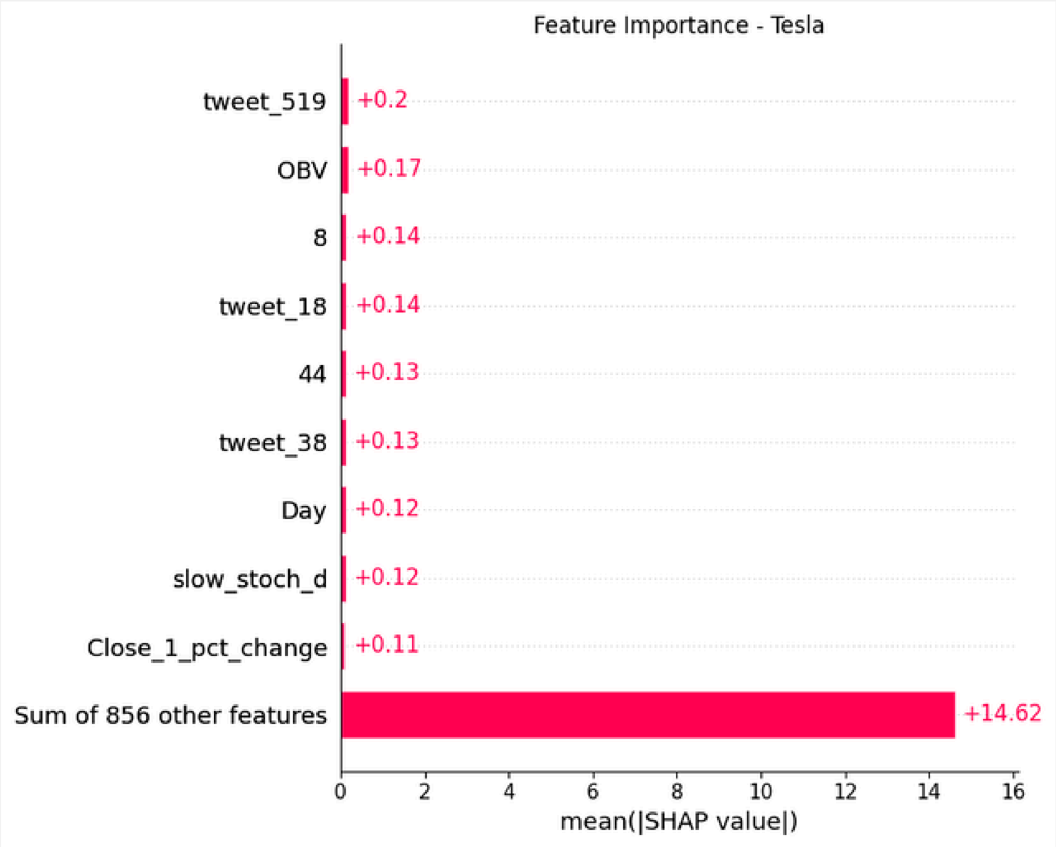
$$Recall = \frac{TP}{TP + FN}$$

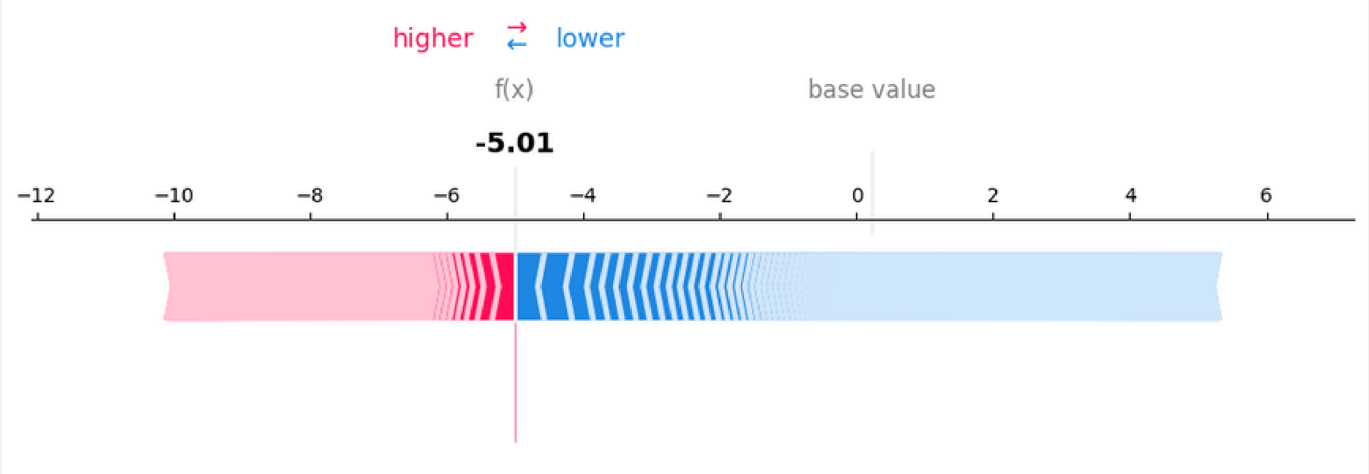
$$F_1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$



# Explainability: SHAP



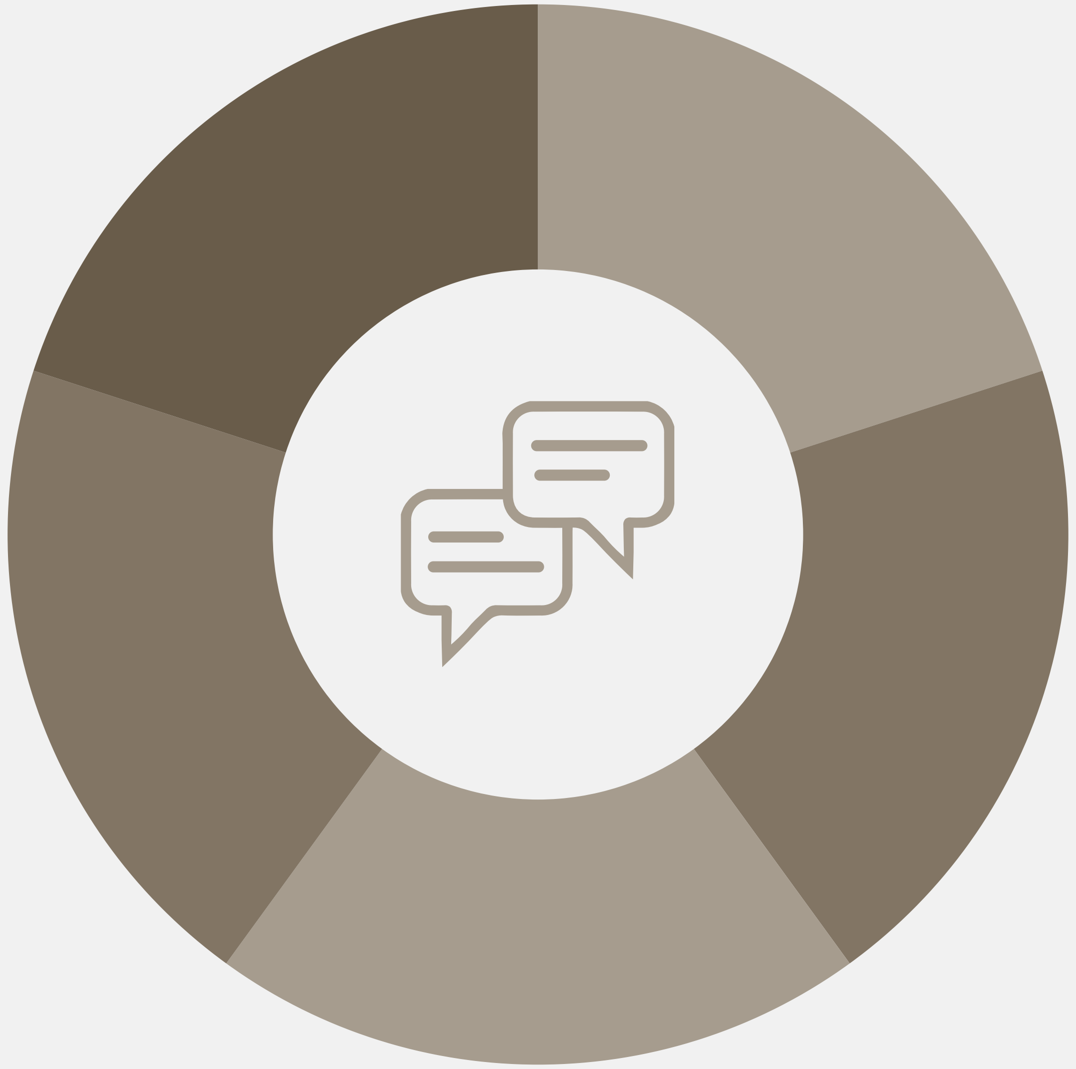



# Experiment Setups

- Setup: Combination of different features
- There are 6 different combinations/Setups
- Each Setup will be evaluated and compared

# Different Feature Combinations

1. Setup 1: The base line model with technical indicators only
2. Setup 2: Technical indicators + tweets (processed via FinBERT)
3. Setup 3: Technical indicators + news (processed via FinBERT)
4. Setup 4: Technical indicators + news (processed via FinBERT + standardised + UMAP-reduced)
5. Setup 5: Technical indicators + tweets (processed via FinBERT) + news (processed via FinBERT)
6. Setup 6: Technical indicators + tweets (processed via FinBERT) + news (processed via FinBERT + standardised + UMAP-reduced)



# Global Setups Comparison

- Individual Setup Performances
- Globally compared with each other
- Performance-based Insights

# Setups Performance

| Configuration | Accuracy | Recall | Precision | F1-Score | ROC AUC Score | F1-Score Gap between Classes |
|---|---|---|---|---|---|---|
| Setup 1 | 54% | 54% | 64% | 56% | 59.09% | 15% |
| Setup 2 | 67.57% | 68% | 64% | 65% | 67.13% | 46% |
| Setup 3 | 47.22% | 47% | 43% | 45% | 33.68% | 53% |
| Setup 4 | 59,46% | 59% | 56% | 57% | 52% | 52% |
| Setup 5 | 67.57% | 68% | 66% | 67% | 55.94% | 38% |
| Setup 6 | 72.97% | 73% | 72% | 72% | 73.43% | 31% |

- Setup 6 outperforms other configurations by a significant amount
- Sentiment with proper pre-processing enhance predictions
- F1-Score gap may exist due to class imbalance
- Worst: Setup 3 → unprocessed news vector introduce noise

# Effect of Extra Pre-processing

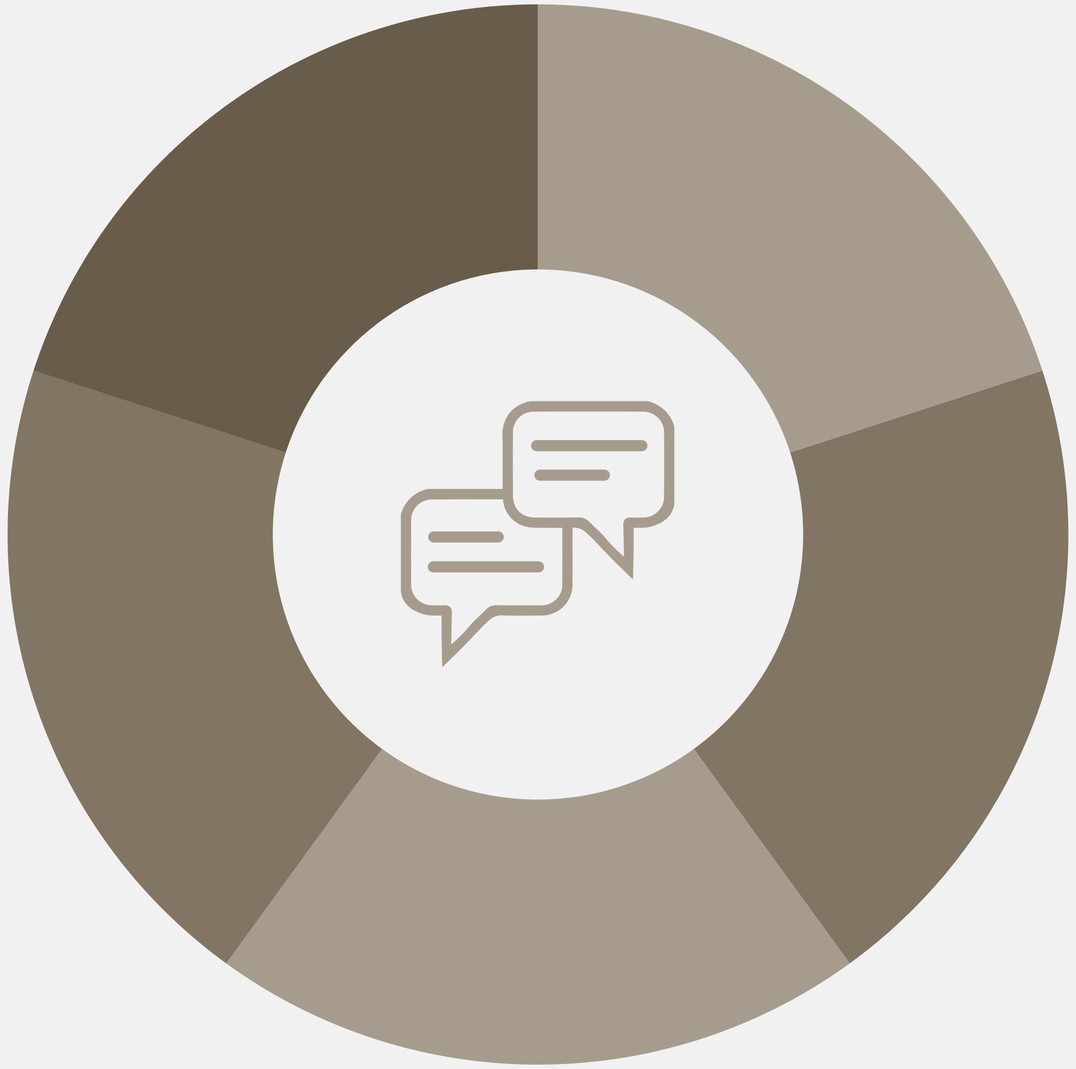
| Metric | Setup 3 | Setup 4 | Improvement |
|---|---|---|---|
| Accuracy | 47.22% | 59.46% | +12.24% |
| F1-Score | 45% | 57% | 12% |
| ROC AUC Score | 33.68% | 52% | +18.32% |
| F1-Score Gap | 53% | 52% | +1% |

| Metric | Setup 5 | Setup 6 | Improvement |
|---|---|---|---|
| Accuracy | 67.57% | 72.97% | +5.4% |
| F1-Score | 67% | 72% | 5% |
| ROC AUC Score | 55.94% | 73% | +17.49% |
| F1-Score Gap | 38% | 31% | 7% |

- Setup 3 & 5: Raw news
- Setup 4 & 6: Extra pre-processed news
- Extra pre-processing: standardisation + UMAP-reduction
- Standardisation + UMAP improved performance dramatically
- Tweets added value, especially when combined with well-pre-processed news data

# Key Insights

- Raw sentiment data (news) adds noise unless transformed
- Extra pre-processing steps (standardisation + UMAP reduction) improve prediction quality and class balance
- Tweets enhance predictive power when added to technical indicators and news



# Feature Impact

- Most influential features across Setups
- Comparative Feature Interpretation
- General Observations & SHAP Insights

# Top Influential Features

| Configuration | Top 5 Most Important Features |
|---|---|
| Setup 1 | 1. slow_stoch_d<br>2. Close_8_pct_change<br>3. Close_1_pct_Change<br>4. Day<br>5. Volume_MA_5 |
| Setup 2 | 1. tweet_519<br>2. slow_stoch_d<br>3. OBV<br>4. tweet_18<br>5. Close_1_pct_change |
| Setup 3 | 1. art_159<br>2. art_151<br>3. art_532<br>4. art_126<br>5. art_540 |
| Setup 4 | 1. Close_1_pct_change<br>2. slow_stoch_d<br>3. Day<br>4. OBV<br>5. Close_5_pct_day |
| Setup 5 | 1. art_303<br>2. art_217<br>3. art_286<br>4. tweet_519<br>5. art_95 |
| Setup 6 | 1. tweet_519<br>2. OBV<br>3. UMAP_8<br>4. tweet_18<br>5. UMAP_44 |

- Text-based features from tweets, namely tweet_519, became important in tweet-augmented models
- Feature engineering brings back technical indicators as top contributors

# Baseline Vs. Tweet Augmentation

| Configuration | Top 5 Most Important Features |
| --- | --- |
| Setup 1 | 1. slow_stoch_d<br>2. Close_8_pct_change<br>3. Close_1_pct_Change<br>4. Day<br>5. Volume_MA_5 |
| Setup 2 | 1. tweet_519<br>2. slow_stoch_d<br>3. OBV<br>4. tweet_18<br>5. Close_1_pct_change |

- With additional tweets variable, the importance was shifted towards text-derived features (tweet_519, tweet_18)
- Some technical indicators still present = compatibility
- Tweets enhance model understanding without neglecting traditional signals

# Raw Vs. Processed News

| | |
|---|---|
| Setup 3 | 1. art_159<br>2. art_151<br>3. art_532<br>4. art_126<br>5. art_540 |
| Setup 4 | 1. Close_1_pct_change<br>2. slow_stoch_d<br>3. Day<br>4. OBV<br>5. Close_5_pct_day |

- Raw unprocessed news dominated the model but performed the worst (lack interpretability & confuse the model)
- Processed news reintroduced technical indicators = complement
- Richer data may worsen model predictive power without proper pre-processing

# Complete Configurations

| | |
|---|---|
| Setup 5 | 1. art_303<br>2. art_217<br>3. art_286<br>4. tweet_519<br>5. art_95 |
| Setup 6 | 1. tweet_519<br>2. OBV<br>3. UMAP_8<br>4. tweet_18<br>5. UMAP_44 |

- Similar as before, unprocessed news was dominating
- Processed news introduced balance between technical indicators, social media, and the processed news itself
- Better feature synergy & model interpretability

# General Observations & SHAP Insights

- Technical Indicators: OBV & Close_1_pct_change indicate robustness
- Tweet-derived features: tweet_519 is significant
- Pre-processed news introduced balance & better model performance
- Based on the SHAP Summary Plot (Beeswarm), slow_stoch_d, one of the most important features, performed more consistently when all features (technical indicators + social media + processed news) are incorporated

# Conclusion

- 📌 Developed a hybrid sentiment-aware forecasting model integrating FinBERT + XGBoost.
- ✅ Demonstrated that financial sentiment (tweets & news) significantly enhances short-term stock movement prediction when properly processed.
- 🔍 Pre-processing (e.g., UMAP + standardisation) improved model balance and interpretability.
- 📈 SHAP results confirmed the complementary value of sentiment and technical indicators.

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# Thank you

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